

Your Paper Title Here

Your Name(s)

Date

Abstract

Write your abstract here.

Keywords

Keyword 1, Keyword 2, Keyword 3, Keyword 4

Introduction / Background

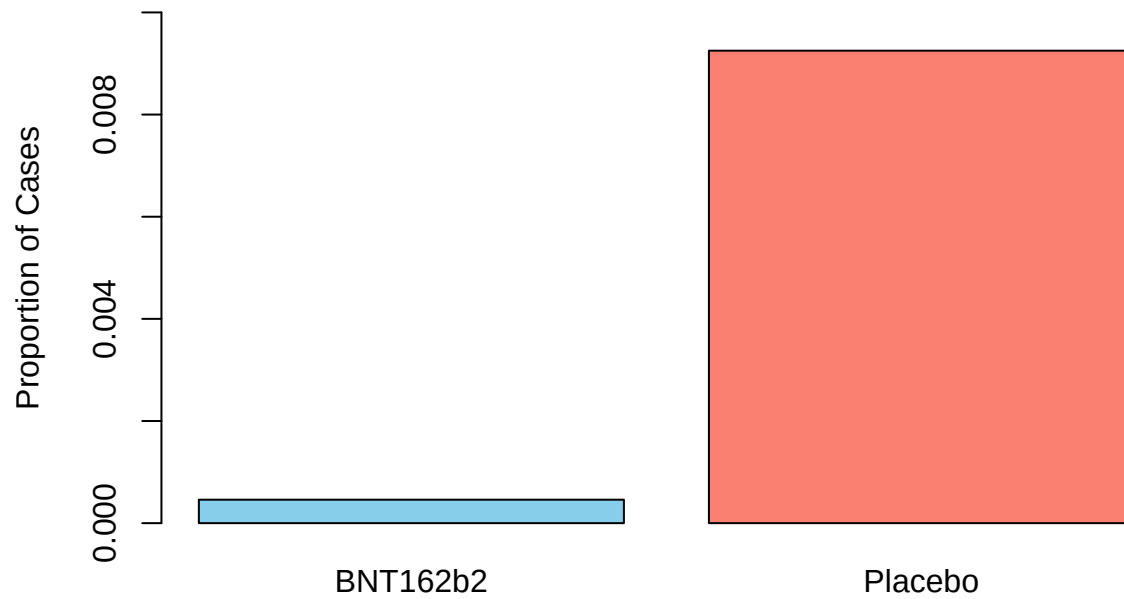
The December 2020 Emergency Use Authorization (EUA) of the BNT162b2 vaccine marked a pivotal shift in the global effort to mitigate the SARS-CoV-2 pandemic. Developed through a collaboration between Pfizer and BioNTech, this two-dose regimen represented a departure from traditional vaccine technologies. As noted in contemporary scientific literature, while viral vector approaches were utilized by manufacturers like Janssen and AstraZeneca, BNT162b2 utilized lipid nanoparticle-formulated nucleoside-modified RNA (mRNA) to induce an immune response against the viral spike protein. To evaluate the safety and clinical impact of this technology, a rigorous, multinational, placebo-controlled efficacy study was conducted and published by Polack et al. (2020). This trial followed a 1:1 randomization ratio, assigning over 34,000 participants to either the vaccine or placebo groups. The study was designed to reach statistical maturity upon the occurrence of 170 laboratory-confirmed COVID-19 cases in participants without prior evidence of infection. The resulting data demonstrated a significant disparity in outcomes. Of the 170 confirmed cases, 8 occurred in the BNT162b2 cohort ($n = 17,411$), while 162 occurred in the placebo cohort ($n = 17,511$). This results in an observed infection rate of approximately 0.046% for the vaccine group compared to 0.925% for the placebo group.

```
trial_matrix <- matrix(c(8, 17411-8, 162, 17511-162),
                      nrow = 2,
                      dimnames = list(Outcome = c("Infected", "Healthy"),
                                       Group = c("BNT162b2", "Placebo")))

rates <- trial_matrix["Infected", ] / colSums(trial_matrix)

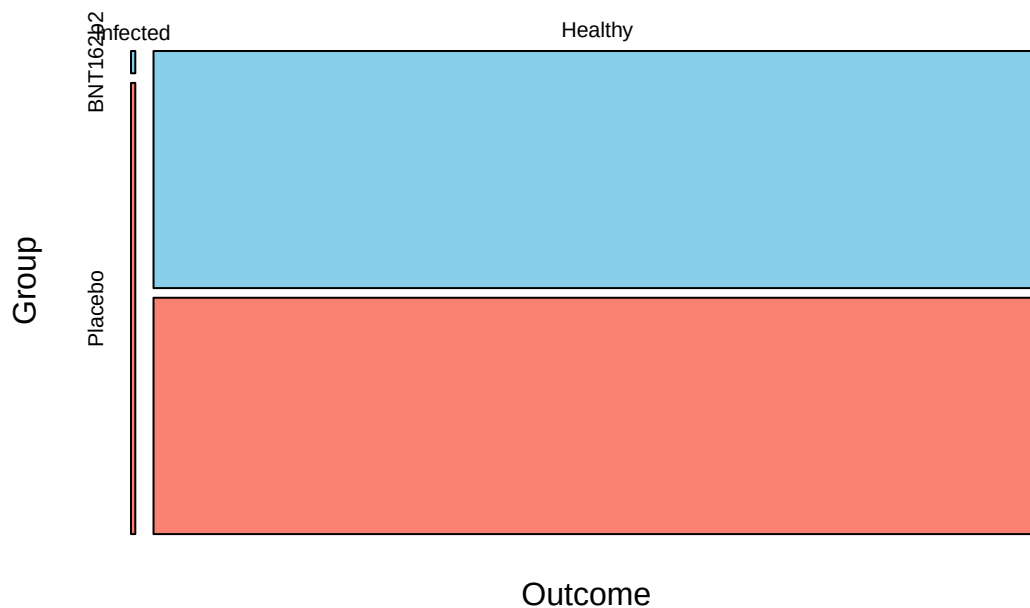
barplot(rates,
        main="Infection Rate by Group",
        ylab="Proportion of Cases",
        col=c("skyblue", "salmon"),
        ylim=c(0, 0.01))
```

Infection Rate by Group



```
array <- as.array(trial_matrix)
mosaicplot(array, main="Mosaic Plot of Distribution", sub = NULL, xlab = NULL, ylab = NULL,
  sort = NULL, off = NULL, dir = NULL,
  color = c("skyblue","salmon"), shade = FALSE, margin = NULL,
  cex.axis = 0.66, las = par("las"), border = NULL)
```

Mosaic Plot of Distribution



This analysis seeks to evaluate the efficacy of the vaccine based on these findings. Specifically, we test whether the vaccine efficacy (ψ) is significantly greater than the 30% threshold required for regulatory approval. We formulate our research goal through the hypotheses $H_0 : \psi = 0.3$ versus $H_1 : \psi > 0.3$.

Statistical Methods

Model

Describe the statistical model used.

Likelihood Inference

Detail the likelihood approach.

Bayesian Inference

The Bayesian approach requires the specification of a prior distribution for π that represents our state of knowledge before observing the trial data. We adopt a conjugate Beta prior, $\pi \sim \text{Beta}(a, b)$, which is mathematically advantageous as it ensures the posterior distribution will also belong to the Beta family. To determine the values of the hyperparameters a and b , we perform a prior elicitation based on a “pessimistic” perspective of vaccine performance. We assume a prior median for π of 0.5, reflecting an initial belief that the vaccine and placebo groups are equally likely to result in an infection. Furthermore, we reflect regulatory skepticism by setting only a 5% prior probability that the vaccine efficacy meets the 30% threshold ($P(\psi \geq 0.30) = 0.05$). Converting this efficacy threshold to the π scale using the transformation $\pi = \frac{1-\psi}{2-\psi}$, we target a distribution where the 5th percentile is approximately 0.4118. Solving these constraints yields the parameters:

$$a = 43.04, \quad b = 43.04$$

The likelihood of observing t infections in the vaccine group out of n total cases is given by the binomial probability mass function:

$$f(t|\pi) = \binom{n}{t} \pi^t (1 - \pi)^{n-t}$$

Applying Bayes’ Theorem, the posterior distribution $f(\pi|t)$ is proportional to the product of the likelihood and the prior. Due to the conjugacy of the Beta-binomial model, the posterior distribution for π is updated by adding the observed successes and failures to the prior parameters:

$$\pi|t \sim \text{Beta}(a + t, b + n - t)$$

With our observed data of $t = 8$ and $n = 170$, the posterior becomes $\pi|t \sim \text{Beta}(51.04, 205.04)$. Following the calculation of this distribution, we transform the results back to the ψ scale. This allows for the determination of the posterior median for vaccine efficacy and a 95% Highest Posterior Density Interval (HPDI) to quantify uncertainty. Finally, the posterior probability $P(\psi > 0.30|t)$ is calculated to evaluate the strength of evidence in favor of the research hypothesis.

```

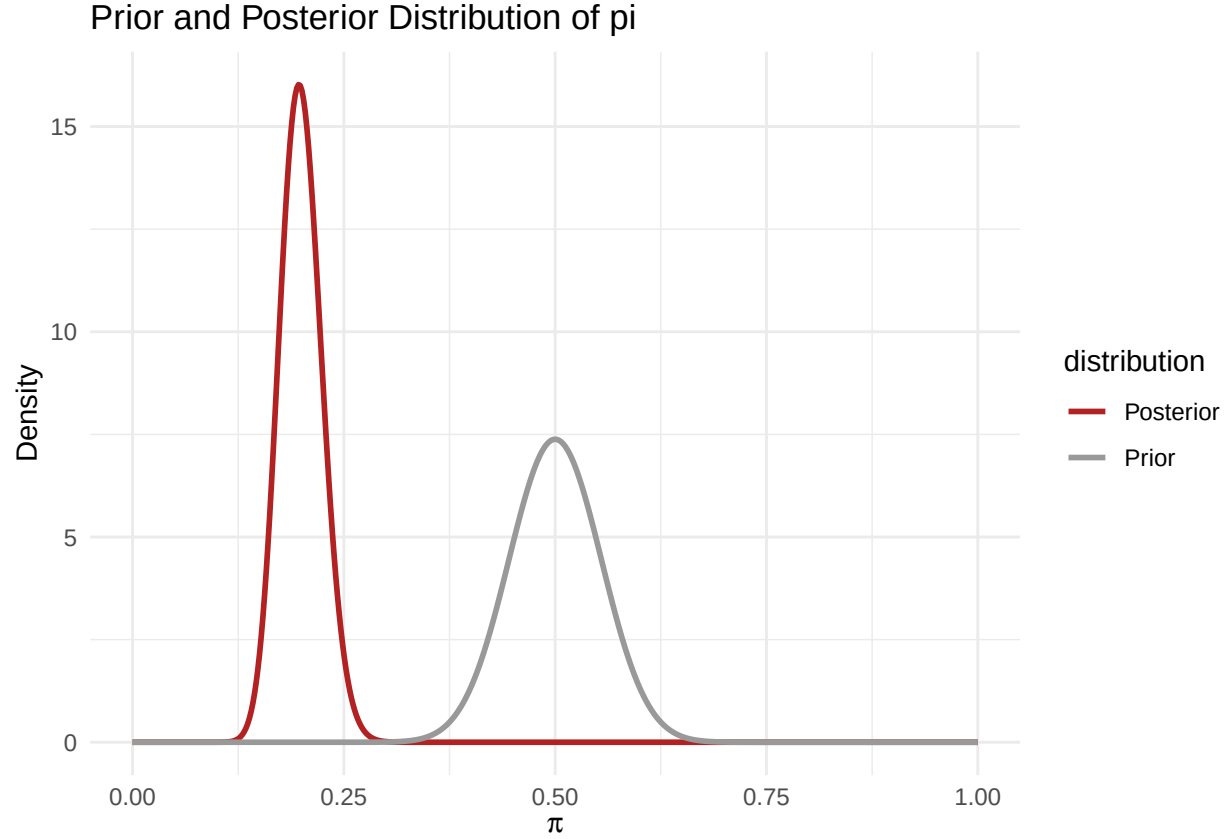
library(ggplot2)

# Define parameters
a_prior <- 43.04
b_prior <- 43.04
t_obs <- 8
n_total <- 170
a_post <- a_prior + t_obs
b_post <- b_prior + (n_total - t_obs)

# Generate data for plotting
x <- seq(0, 1, length.out = 500)
df <- data.frame(
  pi = rep(x, 2),
  pdf = c(dbeta(x, a_prior, b_prior), dbeta(x, a_post, b_post)),
  distribution = rep(c("Prior", "Posterior"), each = 500)
)

# Plot
ggplot(df, aes(x = pi, y = pdf, color = distribution)) +
  geom_line(linewidth = 1) +
  labs(title = "Prior and Posterior Distribution of pi",
       x = expression(pi),
       y = "Density") +
  theme_minimal() +
  scale_color_manual(values = c("Prior" = "gray60", "Posterior" = "firebrick"))

```



Results

The Bayesian analysis provides a probabilistic assessment of the vaccine efficacy parameter, ψ , derived from the posterior distribution of π . Using the pessimistic prior distribution ($Beta(43.04, 43.04)$) and the trial data of 8 vaccine infections out of 170 total cases, the posterior distribution for π was found to be $Beta(51.04, 205.04)$. The posterior median for the probability that an infected individual belonged to the vaccine group, π , is 0.199. When transformed to the vaccine efficacy scale using the relationship $\psi = \frac{1-2\pi}{1-\pi}$, this yields a posterior median vaccine efficacy of 75.1%. The uncertainty in this estimate is characterized by a 95% Highest Posterior Density Interval (HPDI) for ψ of [66.5%, 82.2%]. To evaluate the research hypothesis $H_1 : \psi > 0.3$, the posterior probability was calculated. The analysis indicates that the posterior probability of the vaccine efficacy exceeding the 30% regulatory threshold, $P(\psi > 0.30 | \text{data})$, is greater than 0.9999. These inferential statistics are summarized in Table 2.

Table 2: Bayesian Posterior Summaries for Vaccine Efficacy (ψ) | Metric | Posterior Estimate || :—
 | :— || Posterior Median (ψ) | 75.1% || 95% HPDI (ψ) | [66.5%, 82.2%] || $P(\psi > 0.30 \mid \text{data})$ | >
 0.9999 |

Discussion / Conclusion

Discuss / conclude here.

Bibliography

Brown, B. (2024). *Lecture Title*. Lecture slides, Course Name, University Name.

Doe, J. (2020). Title of the Paper. *Journal Name*, 12(3), 45-67.

Last, F., & Last, F. (2025). *Book Title*. Publisher.

Smith, A., & Johnson, C. (2023). *Title of the Online Article*. Retrieved from <https://www.example.com>.

Appendix

Code

You can reference your code in the appendix (sample here).

Proofs

If applicable, include detailed mathematical derivations or additional theoretical explanations.